SdS

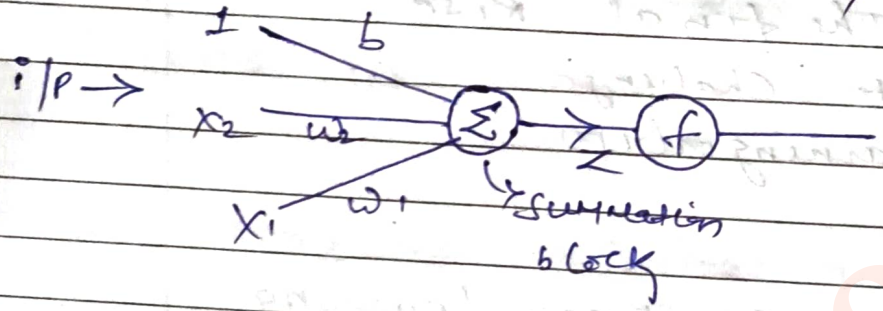
Day - 4

What is perceptron? Perceptron is Neuron

- it is an algorithm used for supervised ML problem.
- it design of algorithm made it the building block of deep learning

Design

Mathematical Model / function / algorithm



(w_1, w_2) → weights | x_1, x_2 = input

(b) → Bias

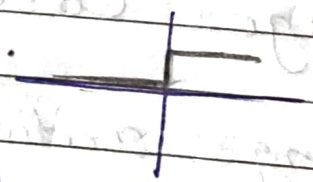
Σ / Z = $w_1 x_1 + w_2 x_2 + b$

f = activation fn used to bring the Z function into a given range (-1, 1)

Step fn used

Relu fn

tanh

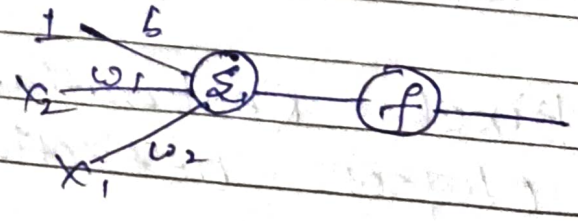


Z > 0 → 1

Z < 0 → 0

eg:

ig	cgpa	placed
78	7.8	1
64	5.1	0



Steps

1) Training
2) Prediction

ShivamXCode01

SDS

x_1 → iq (let $w_1 = 1$, $w_2 = 2$, $b = 3$ (with the help of training)

x_2 → cgpa

eg: 5.1, 100 $100 \times 1 + 5.1 \times 2 + 3 \times 1$

final o/p → 1 placed $100 + 10.2 + 3$

$= 113.2 \rightarrow z > 0$ ①

- for prediction we need the value of weights and bias that we can it with the help of training.

→ As input increase then weight increase.

- Deep learning is heavily inspired by our own nervous system

→ Neuron vs Perceptron

1) Very Complex — simpler than Neuron

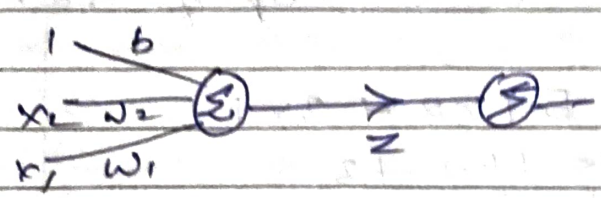
2) processing state is unknown — known (Summation + Activation fn.)

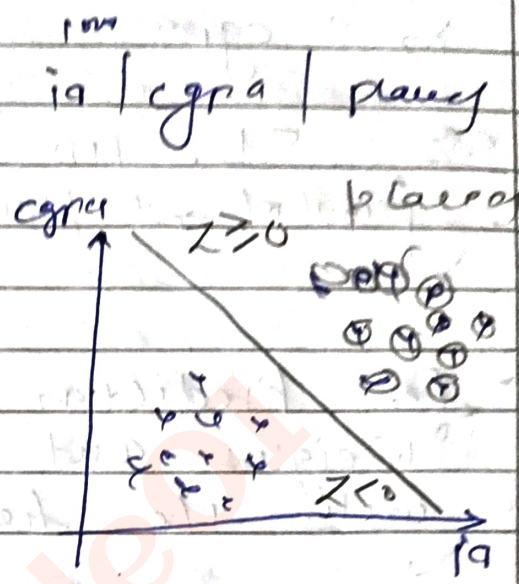
3) Neuroplasticity → Means input can change according to weight — Can't change.

- weights gives the weitage of input on the prediction
- weights tells about the feature importance.

SDS

→ Geometric Intuition





$y = 1, 0 \qquad z = w_1 x_1 + w_2 x_2 + b$

$$y = f(z) = \begin{cases} 0 & z < 0 \\ 1 & z \geq 0 \end{cases}$$

$w_1 = A \quad w_2 = B \quad b = C$

$x_1 = x \quad x_2 = y$

$Ax + By + C \rightarrow$ line eq

$Ax + By + C \geq 0$ → Region

$Ax + By + C < 0$

- perceptron is line that is used create Regions. and divide class in Region.

So,

Perceptron is called Binary Classifier

- it only classify linear/sort of linear data

In non-linear data perceptron fails.

→ Code Demo.